



AIAA 93-0308

NASA's Hypersonic Flight Research Program

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Washington, DC 20546

**31st Aerospace Sciences
Meeting & Exhibit
January 11-14, 1993 / Reno, NV**

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ABSTRACT

An overview of the flight research component of NASA's Generic Hypersonic Program is presented. The program history, philosophy, and rationale are discussed. Flight opportunities are identified for the conduct of the program. Beginning with the results of the Generic Hypersonic Flight Workshop held at NASA Dryden in 1988, the evolution of the Pegasus piggy-back experiments, and the proposed dedicated flights of the Pegasus/SWERVE concept are described. The Pegasus offers a rare opportunity to acquire research data on a piggy-back basis using a winged accelerating vehicle in a portion of the airbreathing corridor. Instrumentation and experiments can be included in the developmental and operational flights of the vehicle. Data can be collected without interfering with the primary objective of the flight. A great deal of hypersonic data can be obtained in this way at low cost. The Pegasus has successfully flown twice yielding excellent flight data. The choice of the Sandia/SWERVE vehicle was based on the fact that it had flown several times and could be very accurately controlled, and that the geometry was highly amenable to computational fluid dynamics (CFD) modeling. Preparations are currently underway to conduct a hypersonic cross-flow transition experiment on a specially designed glove attached to the delta-winged first stage of Pegasus. Descriptions of future piggy-back experiments are also outlined.

I. INTRODUCTION

World-wide interest in hypersonic flight has recently been renewed. A wide range of vehicle concepts, from cruise vehicles and missiles to single-stage-to-orbit (SSTO) and two-stage-to-orbit (TSTO) vehicles, is being considered. These are slender lifting configurations with complex geometries and highly 3-dimensional mean aerodynamic flows. Several characteristics make these vehicles unique and define an era of new hypersonics such as: **airbreathing en-**

gines, long residence times in the airbreathing corridor, highly integrated airframe/propulsion systems, and actively cooled structures. For airframe integrated vehicles, the net thrust may be a small difference between two very large numbers, thus requiring extremely accurate predictive capabilities

These hypersonic vehicles will operate at flight conditions very different from those experienced by subsonic or low supersonic vehicles. Our experience with wind tunnel facilities, computational capacity and flight research has provided the framework for the lower speed areas. This is not the case for airbreathing hypersonic flight, **where verification and development through experiment will not be possible in ground facilities, currently or in the foreseeable future.** The problems of hypersonic flight are primarily associated with the high stagnation temperatures/enthalpies associated with very high speeds. These higher temperatures result in high heat transfer to the vehicles, changes in the gas characteristics around the vehicles, and changes in the chemical constituents of air which can directly affect the classical combustion phenomena. For airbreathing hypersonic vehicles, several key technical and sociological problems are introduced which will be difficult to overcome. These problems include the severe operational environment, the need to totally or partially integrate the propulsion system with the airframe, the ultimate disciplinary coupling of the vehicle subsystems, and the long residence times in an environment at high levels of aerodynamic heating and dynamic pressure.

The high-level multidisciplinary nature of hypersonic vehicles also introduces unique problems in test techniques. Ground facilities are limited in their ability to replicate fluid mechanics phenomena in the hypersonic flight regime. Physical limitations in the facilities manifest themselves in an inability to match many critical parameters such as enthalpy, Reynolds number, chemical composition and dynamics, introduction of secondary flows, and duplication of wall conditions. Consequently, correction factors must be applied to whatever data are obtained in ground test facilities, yet many of these critical parameters can only be obtained in the real environment of flight.

Computational fluid dynamics (CFD) is evolving as the long term solution to some of the limitations of ground test facilities and will permit predictions to be made that are beyond the capabilities of ground facilities. The process of developing CFD codes requires either an explicit modeling of

the physics of hypersonic flow or an empirical process by which the codes are calibrated. In order to accomplish either of these, a reference set of data must be obtained. The fluid mechanical behavior of the vehicle must be measured in actual hypersonic flight in order to validate computational methods, calibrate codes, and define correction methods for ground test results.

The limitations of ground capabilities also manifest themselves in the other disciplines. For example, advanced materials and structural concepts cannot be fully tested in any ground facility due to the inability to accurately simulate the combined flight conditions to which the structure will be subjected. The current process is to investigate a few of the important aspects of the advanced material or structure at a time and in a particular facility, then hopefully, sum the results from a number of such investigations to get a sufficient prediction of the material or structures behavior. While a flight research enhancement to this process will not fully answer all of the questions for all advanced technologies, flight testing will provide the opportunity to validate new technologies in the real environment.

Thus a critical void exists in the traditional design validation process that needs to be filled. The design process itself is further complicated by an unprecedented coupling of the various aerospace disciplines. An appropriate approach to the safe and efficient design of the above vehicles mandates the use of sub-scale, fundamental flight experiments and technology demonstrators, in addition to ground test evaluation and theoretical analysis (computer simulation). This is the so-called three-legged stool. It is an approach for reducing the existing large uncertainties of current computational design until the development of complex theoretical and experimental methods are brought to a satisfactory, validated predictive capability. Flight test, when performed correctly in the actual environment, establishes the true measure of performance; whereas, ground-test and simulation attempt to predict performance. It must be mentioned, however, that flight test also has its limitations and uncertainties.

Perhaps most critical to maturing the technology of hypersonic flight, are the integrated systems issues. Single discipline research is definitely essential; however, real vehicles are integrated systems that depend upon a number of advanced, interdisciplinary technologies to correctly function. An example of these integration issues is actively cooled structures where the manufacturing process, materials, structural concept, cooling medium, wall temperatures, fluid mechanics, and

flight conditions are all factors in establishing performance characteristics. Research on these integrated vehicle issues can only be addressed completely in a flight vehicle. Clearly then, flight should be an essential part of the national hypersonics research program.

The renewed interest in lifting, slender-body hypersonic, airbreathing vehicle configurations in the operational range of Mach numbers from 5 to 25 and altitudes from 50K-200K feet (at high dynamic pressure) mandates some major advances in the hypersonics state of the art. These new vehicles consist of configurations that have small leading edge radius (pointed) and highly 3-dimensional (complex) geometries. The aerodynamics of such vehicles are rich in viscous interaction phenomena over both the vehicle scale and also over the much shorter length scales. Other attributes include very high levels of aerodynamic heating and, in the case of propulsion-integrated vehicles, small thrust-minus-drag values resulting in extreme performance sensitivity. The propulsion-airframe integration problem cannot be separated from the exterior aerodynamics of these hypersonic vehicles and, as a result, vehicle lift, drag, thrust, and moments are all highly coupled.

A report entitled "Review of Aeronautical Wind Tunnel Facilities" (NRC, 1988), chaired by Dr. Smelt, provided strong advocacy for the continuing and essential role of flight testing in advancing the hypersonic state of the art. In particular, four significant benefits can accrue from flight testing: (1) validation/calibration of ground facilities and test-data, thus establishing scaling relationships between ground-test and flight test; (2) establishing conditions of flow quality and instrumentation precision requirements for ground testing directly from the availability of actual in-flight data; (3) validation of subsystem design concepts; and (4) providing data for the development/validation of CFD codes. It is the latter, in particular, that provided the impetus for the 1988 flight test workshop. It is inevitable that CFD will play an even more visible role in hypersonics than in previous applications and its usage will increase substantially. One major challenge, in the present context, will lie in the requirement for higher computational accuracy for CFD and, ultimately, that can only be accomplished through validation in the real flight environment.

II. NASA'S GENERIC HYPERSONIC PROGRAM SUMMARY

In 1988, NASA initiated a long-term investment in hypersonics that addressed a broad, more fun-

damental set of technology needs that would enable and enhance future, slender-vehicle, hypersonic missions. The NASA Generic Hypersonic program (GHP), which resides in the Office of Aeronautics, was initiated in 1988 and was reviewed by the USAF Scientific Advisory Board in June of 1991 in Washington, DC. It is a long-range, multidisciplinary endeavor that stressed fundamental understanding of the controlling hypersonic physical processes and promoted new hypersonic technology ideas. The program emphasizes research and technology development focusing primarily on slender, airbreathing hypersonic vehicles that use highly integrated airframe/propulsion concepts. The scope is quite broad and included various forms of accelerator and cruise vehicles, space and planetary vehicles, and hydrocarbon and hydrogen fueled hypersonic vehicle systems. The GHP also stresses technical issues rather than committing to specific missions and it advanced high pay-off hypersonic technologies. Key technical problems are identified in the areas of hypersonic boundary layers (especially turbulent) and high-speed transition, turbulence modeling, mixing and combustion at supersonic speed, real gas effects at high temperature, rarefied flows at high altitude, and high specific strength metal-matrix composite materials. The primary objective of the GHP is to provide the capability to respond to future NASA, DoD and DoE broad mission and application needs. The GHP has continued to provide a sustained NASA in-house research effort and is strongly supported by academia and the aerospace industry. Cost-effective partnerships with industry have been initiated to streamline the transfer of technical results for the design and development of future aerospace vehicles. Participation by academia continues to involve NASA-sponsored, cooperative research and the rededication of aerospace curricula to hypersonics studies and understanding.

Research is conducted in the classical disciplines of hypersonic gas dynamics, airbreathing propulsion, structures and materials, and flight systems. Code validation is an area receiving extraordinary emphasis by the GHP and the program will advance technology in these areas using unique NASA facilities and a variety of computational capabilities. The program enhances the research in the classical disciplines with new efforts in the areas of technology integration research, advanced instrumentation research, facilities research, and **flight research**. These "new" elements are regarded as integral to all of the classical disciplines. An important focus of the GHP is the enhancement of overall hypersonic capability through activities in

ground-based facilities research, flight experimentation, development of advanced non-intrusive flow-field instrumentation, multidisciplinary research addressing the highly coupled complex interaction of integrated components, and development of CFD algorithms of unprecedented accuracy.

Two new areas of development were proposed in recent years: (1) **High Mach Number Flight Research** which builds on the ongoing successful Pegasus piggy-back experiment program, and (2) ground-based **Critical Technology Development**. Both were designed to address several of the following activities:

- analytical and experimental studies into novel hypervelocity true enthalpy ground facilities capable of testing large-scale models (such as large ballistic range, liquid-air arcjet, and high power arcjets)
- quiet tunnel technology for higher Mach and Reynolds numbers
- benchmark assessor experiments and prototype pilot facility development
- advances in hydrocarbon fuel technology and applications, other innovative propulsion concepts
- advanced, high specific-strength, airframe and engine materials
- hypersonic flight research using SWERVE/Pegasus or equivalent system to meet the high Mach number real gas aerodynamic needs that cannot be met with existing ground facilities.

Flight experiments were to be conducted to investigate select phenomena that are beyond the current ground-test capability, to clarify physical phenomena not fully simulated in current ground facilities, flight-validate advanced instrumentation, and complement ground and numerical experiments. These flight experiments would serve to provide: (1) early flight validation of the numerical (CFD) codes that have been used extensively for the design evaluation of highly instrumented, flight test-bed vehicles operating in the Mach 4-25 test range; (2) a generic technology base for the design of future, slender, hypersonic vehicles; and (3) scientific data that may not be available from the anticipated National Aero-Space Plane (NASP) flights.

III. GENERIC HYPERSONIC FLIGHT RESEARCH PROGRAM

NASA's Generic Hypersonic flight research program was defined in June 1988 at the Flight Workshop held at the Dryden Flight Research Facility (DFRF). This workshop was attended

by representatives from NASA, other government agencies, industry, and academia. A roadmap for hypersonic flight research similar to the one shown in figure 1 was developed at this workshop. The conclusions reached during the workshop regarding hypersonic flight testing are presented in Appendix I. The overall objectives of the program were to conduct flight experiments to:

- Provide low-cost, near term flight test programs for validating and integrating critical hypersonic technologies.
- Validate critical hypersonic technologies or disciplines that are beyond the capability of current ground facilities.
- Develop the instrumentation systems required to operate and perform experiments on hypersonic vehicles. The instrumentation systems should be required to provide quality test information in the severe hypersonic environment.
- Provide fundamental experimental information for future hypersonic research

The proposed approach was to: (1) identify flight test needs; (2) define and develop the objectives and technical requirements; (3) develop supporting ground facility and theoretical studies; and (4) develop a flight test-bed that would be capable of satisfying the technical needs. These proposed objectives and the approach continue to be the major emphasis of the flight program.

IV. MAJOR HYPERSONIC ISSUES

Hypersonic vehicle designs require knowledge of the location and extent of transitional flows and any relaminarization. Boundary layer transition (such as departing from and returning to laminar flow) significantly impacts the aerodynamics (skin friction drag and control effectiveness), the aeroheating (peak rate and integrated load), materials (placement and sizing), and propulsion performance. Fundamental experiments need to be conducted over ranges of Reynolds numbers, Mach numbers, angles of attack, bluntness, curvature, roughness, waviness, pressure gradients, and shock waves. Both Pegasus piggy-back and SWERVE/Pegasus programs have defined transition experiments. These will be discussed in detail later.

The influence of stream molecular contaminants and fluctuation fields on scramjet combustor performance is an extremely important flow quality issue. Many existing scramjet test facilities have freestream molecular contaminants and disturbance fields which can have a first-order influence on scramjet combustor performance. The

processes potentially affected include reaction initiation, reaction rate(s), shear-layer instability, transition, and turbulent mixing rate. Except for radiated acoustics, molecular contaminants and disturbance fields are not present during flight, and quantifications of their significance to combustor performance do not exist. Experience indicates that these ground-facility phenomenon can erroneously enhance combustor performance by as much as 30 percent. Sources of these contaminants and disturbances are primarily heating processes including vitiation, arcjets, heater beds, and associated enthalpy fluctuation fields. Instrumentation must be developed to measure stream contaminant and fluctuation field levels in various scramjet test facilities. Combustor performance data obtained in each facility should then be compared with one another and with the results of a free-flight mixing experiment. The latter being absolutely essential to validate the real environmental conditions.

In addition to the flow quality issue, there are major uncertainties with validity of the ground test results and techniques. These uncertainties arise from a combination of scaling factors, lack of ground simulation capability, and geometrically-complex planforms with many integrated system issues. Single discipline research is of course, absolutely essential; however, real vehicles are integrated systems that depend upon a number of advanced technologies to correctly function. A typical example of the integrated systems issue is actively-cooled structures, where the manufacturing process, materials, structural concept, cooling medium, wall temperatures, fluid mechanics, and flight condition are factors in establishing performance characteristics. For real vehicle applications, it is clear that test methodology and design methodology are intimately related to new facility concepts which address the multidisciplinary nature of these issues. Presently, experimental research on these integrated vehicle issues can only be addressed with a flight vehicle.

There are some very special needs for computational method validation which will be an essential element for the design of hypersonic vehicles. Illustrative of the problem is scramjet engine qualification and certification concerns presented below:

- scramjet concept not proven in flight in the USA
- limited ground test performed to date including limited wind tunnel tests with small models, maximum wind tunnel test Mach numbers below 8 with short duration test times, Reynolds numbers and real gas ef-

- effects not accurately simulated, boundary layer and forebody simulations inadequate
- physics/thermochemistry of the combustor not well understood
- CFD analysis tools not well validated and limited in application
- questions on combustion process, efficiencies, flow path, influence of facility-generated disturbances, and facility-induced contaminants

Hypersonic vehicles (airbreathing, in particular) should be tested at Mach numbers from 5 to 20, at low altitudes (70K to 150K feet), and at high dynamic pressures (q) of 500 to 2,500 pounds per square foot. For airbreathing propulsion, ground facilities can test to Mach numbers of approximately 8 at lower dynamic pressures which represents higher altitudes. Scramjet propulsion requires testing at the same conditions identified for the hypersonic vehicles listed above to better understand its function and performance. These tests should include studies of the flight performance of these engines over a wide Mach number range as well as the influence of external vehicle flowfield and maneuvering effects on engine performance.

The advantages of an airborne flight capability are:

- real atmosphere (non-vitiated air) provides real gas effects, true Reynolds number, realistic flow enthalpies.
- potential of realistic engine geometry and size, forebody operating characteristics and conditions, real data for correlation
- validation of analysis and design tools and aerothermodynamic codes and models
- understanding of flowpath physics and thermochemical processes and engine cycle performance potential
- direct wind tunnel-to-flight comparison with same engine

Several examples of instances where actual flight testing has shown wide variance with wind tunnel and CFD results exist. The Cone Boundary Layer Transition Program is an example of an experiment which has flown on a F-15 aircraft and the results compared with test results obtained in noisy and quiet supersonic wind tunnels (ref. 1). Several other examples may be found in the experiences encountered in the first Space Shuttle flights (ref. 2). A properly designed flight test with appropriate instrumentation, flown at conditions which closely match those in ground facilities, can provide a critical validation of the ground test results. This data would also provide an important validation or calibration of

CFD, provided the conditions of flight are well defined and the instrumentation has high accuracy. Sandia National Laboratories has compared the aerodynamic and flowfield predictions of various CFD codes with the extensive ground test and flight test data base for a hypersonic maneuvering reentry vehicle (ref. 3). Data from over 6000 hours of wind tunnel tests, CFD predictions, and 3 flight tests are available for comparison. Considerable modification and fine tuning of the CFD codes was required to get correlation with pressure measurements and overall aerodynamic coefficients. An adequate aerodynamic heating correlation was achieved only after forcing the CFD analysis to follow the observed flight transition from laminar to turbulent conditions. For risk reduction, safety, and cost reasons, a subscale flight research capability must be available to support hypersonic programs. This capability is especially desired for supporting the development of man rated systems which require high standards of reliability and safety.

Historically, there have been two major arguments against an extensive, structured launch capability, risk and cost. Catastrophic failure and loss of the flight vehicle and test item are often cited as major objections against flight testing. The second argument is cost, the launch and flight vehicles are costly to build and operate. However, the technical need to conduct flight tests (experimentation, validation, and development) to acquire flight data under real world conditions overrides these arguments. Since flight experimentation is a relatively expensive business, its urgent need in the hypersonic arena might result in foregoing some other important research. Therefore, only those experiments that provide data otherwise unobtainable, such as transition, combustion length, and code validation should be flown.

From these major hypersonic issues, the following requirements for hypersonic flight testing may be identified:

- **benchmark experiments on high Mach number boundary layer transition including the effects of adverse pressure gradients, crossflow, and real gas effects**
- **effects of ground facility generated disturbances on mixing (with chemical reactions) in scramjet combustor flowfields**
- **external performance of scramjets including propulsion/airframe integration and installed performance**
- **code validation particularly at Mach numbers above 8**

V. SUMMARY OF EXPERIMENTS

A list of the most important experiments identified during the 1988 Flight Test Workshop is presented below. These experiments were arrived at by consensus through discussion between NASA, industry, academia and other government agencies. These experiments were based upon the use of the Pegasus and SWERVE vehicles as the primary test beds.

- **Boundary Layer Transition**

Experiment (Tollmien-Schlichting (T-S) wave, crossflow, pressure gradient and combinations with pressure gradients)

- **Turbulence Studies**, shock/boundary layer interaction
- **Cowl Lip Heating Experiment**
- **Inlet Technology Flow-Through/Cold Flow Experiment**
- **Vehicle Aerodynamic Experiments**
- **Flush Air Data System Experiment**
- **Scramjet Inlet Operability and Performance Determination**
- **Very-High Temperature Ceramic Leading-Edge and Nosetip**
- **Surface Catalysis Experiment**
- **Flight Performance of Advanced Thermal Protection Materials**

The following paragraphs provide summaries of each of these proposed experiments including specific details concerning the flight conditions and hypersonic configuration limitations that may apply.

Boundary Layer Transition Experiment OBJECTIVE

To study inflight transition of high Mach number boundary layers subjected to various isolated instability modes and combinations. Validate models used to predict these processes.

APPROACH

Unique combinations of flight condition, angle of attack, and test fixture design will be used to generate boundary layer transition in the following modes and combinations:

- TS instability, no pressure gradient
- crossflow instability, no pressure gradient
- TS instability with pressure gradient
- crossflow instability with pressure gradient

The onset and end of transition of boundary layer will be measured. The frequency of the boundary layer disturbances at the surface and in the freestream flowfield will be measured for the purposes of analytical modeling of the transition process and to confirm that the desired transition modes have been achieved.

VEHICLE CONFIGURATION

- Cases without pressure gradient must have

no deviation from the conical surface upstream or in the vicinity of the planned transition regime.

- Cases with pressure gradient must have surface modifications to generate required flowfield. For the pressure gradient without crossflow, this may be achieved with a ramp although, a preferred method is to have a flared bi-conic shape. For the pressure gradient with crossflow case, the flared bi-conic shape is required.
- Nose tip must be sharp enough to insure that the desired transition instabilities are dominant and that the boundary layer is dominated by real gas constituents during transition process. As a minimum, the onset of transition should be about 100 nose radii downstream of the nose to allow swallowing of the nose bluntness effects.
- Some outgassing on nosetip is allowable, no shape change is allowed.
- No outgassing from conical surface ahead of transition is allowed.
- Smoothness of the upstream surfaces should be consistent with a critical Reynolds number of 100 (at an altitude of 85K feet and a Mach number of 14, this corresponds to protuberance of about 0.01 in.).

FLIGHT CONDITIONS

- free stream target Mach number must be greater than 14, slight variation around target Mach number is desirable
- altitude based on local Reynolds number requirement due to combination of nose bluntness and vehicle length
- time at flight conditions to be adequate for spectral analysis instrumentation and Mach number variation
- angle of attack sweeps for crossflow modes and zero angle of attack for other modes

INSTRUMENTATION

- alpha measured to 0.1 degree accuracy
- thermocouples
- calorimeters
- low and high frequency pressure transducers
- acoustic sensors
- boundary layer pop-up rakes with both low frequency and high frequency sensors
- freestream turbulence sensors (may be possible with a support aircraft at 80K feet altitude, or by deploying a sensor package from the aft end of the research vehicle)
- onboard processing for spectral analysis of high frequency data (up to 6 spectra/second or higher)

ISSUES

- Need to determine suitability of ramps for pressure gradient studies of boundary layer

transition.

- Need to determine the effect of gaps, seals, and roughness associated with deployed surfaces.
- Need to improve ability to predict transition location as required to support experiment design.

Turbulence Studies, Shock/Boundary Layer Interaction

JUSTIFICATION

Compressible turbulence modeling is the pacing development of CFD. Modeling is currently based on ground test data bases alone. No flight data exist to validate modeling and extrapolation techniques.

OBJECTIVE

Obtain turbulence modeling validation data at flight Mach numbers and Reynolds numbers representative of airbreathing propulsion vehicles.

APPROACH

Measure pressure, heat transfer, and limited flow field data on hypersonic vehicle adapted with a compression ramp, generic flow-through inlet configuration in turbulent boundary layer flow. A ground test experiment for the same geometry should be performed in the Ames 3.5 foot wind tunnel prior to the flight.

VEHICLE CONFIGURATION

- Two wedge shaped fins with sharp, cooled leading edges are required on the conical surface.
- Some outgassing on the nose is allowable.
- No outgassing on the conical and ramp surfaces upstream of the test fixtures is allowed.
- Fin fixtures must be located in fully developed turbulent boundary layer flow (aft location on vehicle provided there is no interference from the vehicle control surfaces).
- Fins may be deployed at flight condition and retracted after flight data obtained.
- Variable geometry of the fins during the flight test may be required.
- A minimum of four pop-up, 3-hole pitot rake fixtures (15 measurements each) is required. Two on the cone surface outside of fins; two on the cone surface between fins (as shown on figure 6)

FLIGHT CONDITIONS

- conditions that can be fully simulated in ground test facility (Mach < 8)
- very high enthalpy, real gas conditions with turbulent boundary layer, if possible (Mach = 10, 13, and 15)
- altitudes as required for fully developed turbulent boundary layers at test location (Reynolds number Theta/Mach number = 150, transition criteria and transition length

= 1/2 laminar length)

- time at flight condition to obtain variable geometry sweeps, heating, and pressure data (approximately 1-30 seconds)

INSTRUMENTATION

- surface: pressure (80), heat transfer and temperature (20)
- above surface measurements: four rakes with 15 measurements per rake (3-hole pitot pressures), and total temperatures

ISSUES

- Will there be adequate turbulent boundary layer flow on the vehicle?
- Are there devices that can be used to increase the rate of transition and still provide a suitable boundary layer for modeling studies?
- Can small fin radius be maintained in presence of high heating?

Cowl Lip Heating

JUSTIFICATION

Flight experiments will provide the first shock-on-lip experimental data under high Mach number flight conditions. This data is not available from ground based facilities. Flight experiments will provide shock/shock interaction data that are free from tunnel noise and flow unsteadiness. Inability to obtain the flight experiments means uncertainty in predicted flight shock/shock heat flux which may reduce the ability to achieve the desired scramjet performance. Current predictions are validated wind tunnel data at thermally perfect gas conditions and lower dynamic pressures. Future vehicle designs which will encounter shock/shock heating could either be over-designed or possibly fail due to under-design.

OBJECTIVE

Obtain detailed surface measurements of heat flux and pressure on a cylindrical leading edge fin in flows representative of the NASP shock-on-lip problem. In particular, the leading edge radius and freestream Mach number must represent the real gas, finite rate chemistry problem. If possible, the leading edge radius, Mach number, and altitude must represent the non-continuum effects and flight conditions. Quality of the measurements must be adequate for development and validation of CFD methods.

APPROACH

Two optional approaches are proposed:

- The angle of attack of the vehicle is adjusted in order to provide adequate shock strength. The test specimen is deployed on the windward side into the vicinity of the bow-shock, the extension of the specimen is adjusted slightly in order to sweep the resulting sl interaction pattern and measurements are made. Upon completion of the measure-

ments, the specimen is retracted below the skin surface.

- The vehicle is flown at zero angle of attack, the test specimen and a shock generator are deployed into the shock layer, the extension of either the specimen or the shock generator are varied in order to sweep the resulting shock pattern and make flight measurements. The specimen is retracted below the skin surface after completion of the measurements.

VEHICLE CONFIGURATION

- The test fixture(s) (specimens) must be deployed to a position at least outside of the cone boundary layer. The cone boundary layer may have ablation products. There can be no deviations from the cone geometry ahead of the test fixture(s). The fuselage station location of the fixture(s) is affected by the following considerations:
 - far enough aft so that the local cone diameter is adequate to support the experiment
 - far enough forward so that the fixtures do not have to extend excessively to clear the boundary layer
- The test specimen should be a flat surface with a cylindrical leading edge. A leading edge radius of 0.1 in. will fully-simulate the desired NASP configuration. If larger radius is used, it must still be small enough to allow significant levels of finite rate chemistry.
- The width of the specimen must be adequate to provide 2-dimensional conditions near the center. A ratio of width to leading edge radius of 6 has been used in wind tunnel tests.
- If the bow shock option is used, the specimen may need to be curved to match the shock shape to some degree.
- If the shock generator option is used, the shock generator must attempt to correct for the conical shock layer flow to provide 2-dimensional flow on the specimen. The shock generator should be a wedge of about 6 degrees. The separation between the wedge and the test specimen must be long enough to prevent an unstart condition and short enough to prevent interference from expansion waves aft of the wedge.
- The test specimen should not be actively cooled (in the measurement region) to allow thin film heat flux measurements

FLIGHT CONDITIONS

- for a 0.1 inch radius leading edge, Mach = 15 and dynamic pressures of 2100 pounds/square foot (non-continuum effects cannot be simulated for larger radius lips except at higher altitude)
- angle of attack may be required for bow-shock option
- adequate time at flight condition for heat

flux measurement (time derivatives of temperature) at several shock interaction configurations (obtained by variable geometry)

INSTRUMENTATION

- dense distributions of surface pressures and heat flux around the circumference of the test specimen (measurement of heat flux per degree)
- measurement of freestream, undisturbed shock layer flow properties

Inlet Technology Flow-Through/Cold Flow Experiment

JUSTIFICATION

The cowl shock, inlet shoulder, and boundary layer interaction plays a critical role in inlet operability. Large separation or unstart can occur unless the shock is controlled and positioned correctly. The design process for this system must be calibrated and verified. Flight data will provide an increased confidence level in the high speed inlet process. The risk associated with inlet design would be reduced if flight data were available.

OBJECTIVE

Obtain inlet aerodynamics that verifies test data and CFD on cowl shock, inlet shoulder, and boundary layer interaction.

APPROACH

Design a high speed inlet using state-of-the-art CFD codes. Fabricate and ground test inlet models (perhaps up to full scale). Predict shock wave interactions and inlet flow fields using CFD analyses. Fabricate and flight test inlet configuration based upon designs determined from CFD and ground tests. Determine code deficiencies and improve CFD and test techniques.

VEHICLE CONFIGURATION

- Flight inlet that is deployed from surface during testing.
- Test largest scale feasible (sizing criteria based on ratio of boundary layer thickness at cowl lip to inlet height of 5).
- Variable geometry of inlet cowl is required.
- No ablation on the conical surface upstream of the inlet is allowed.

FLIGHT CONDITIONS

- flight Mach numbers between 2 and 10 (or greater)
- for a fixed geometry inlet, zero angle-of-attack and vary Mach number to sweep cowl lip shock wave across body side shoulder
- for variable geometry inlet, constant Mach number and zero angle-of-attack to sweep cowl lip shock wave across body side shoulder
- reasonable test time of 5-10 seconds
- near 2-dimensional, uniform flow entering the inlet during test time
- fully turbulent boundary layer on body sur-

face

INSTRUMENTATION

- two pop-up and retractable pressure and temperature rakes at each location of inlet entrance, throat, and exit
- 50 to 75 static pressures on body-side and cowl surfaces

ISSUES

- Flight experiment must be designed to assure survivability of the inlet.
- Can 2-Dimensional flow be maintained into the inlet during the test time?
- Will flight vehicle control surfaces disturbances be minimized?

Vehicle Aerodynamics

JUSTIFICATION

The level of optimization that can be attempted for an advanced highly-integrated hypersonic vehicle such as the NASP is dependent on the reliability of the aerodynamic design tools. Flight research with a large, well-instrumented configuration is the only means of evaluating the reliability of total vehicle aerodynamic predictions at high enthalpies (with undisturbed freestream air flows). If a confidence level can be quantified with flight data, then the required design margins and risk factors for a NASP-type design can be significantly reduced.

OBJECTIVE

Provide direct comparisons of flight measured and state-of-the-art experimental and theoretical vehicle aerodynamic parameters including performance and stability and control parameters. Determine the component contributions to vehicle lift and drag from the integration of surface pressures. Identify the level of accuracy of the prediction methodologies in the design of future highly optimized hypersonic configurations such as the NASP. Develop and optimize flight test techniques for the determination of total vehicle aerodynamic parameters which are effective in terms of accuracy and use of flight time

APPROACH

Forces and moments will be determined throughout the flight trajectory from static force-balancing methods. Derivatives will be determined at specific flight conditions using dynamic maneuver analysis. Incongruities between flight and theory or wind tunnel data will be resolved when possible through the use of pressure data and improvements to the prediction methods will be accomplished.

VEHICLE CONFIGURATION

- Slender-body, hypersonic configuration with complex geometry and control surfaces to provide precise control of flight conditions.

FLIGHT CONDITIONS

- low amplitude perturbations for the extrac-

tion of stability and control data

- slow angle-of-attack variations of about 2 degree per second for performance studies at a variety of flight conditions
- test conditions will be selected to cover conditions which can be simulated in wind tunnel facilities including conditions which emphasize real gas effects and rarefied flow effects

INSTRUMENTATION

- inertial data (angular and linear rates and accelerations)
- control surface positions
- vehicle mass properties
- freestream air data
- approximately 200 surface pressures distributed on forebody, wings, and base

Flush Air Data System

JUSTIFICATION

The use of a Flush Air Data Sensor (FADS) is expected to be critical to the operation of advanced highly-integrated hypersonic vehicles such as the NASP. The FADS approach has the potential to provide reliable, fault-tolerant measurement of flight conditions without exposing probes or sensors to the hostile hypersonic environment. Real-time FADS measurements may be needed for engine and airframe control system feedback of hypersonic air-breathing vehicles.

OBJECTIVE

This experiment will develop and validate a pressure-based air data sensing system capable of sensing Mach number, altitude, angle-of-attack, and angle-of-sideslip with research quality accuracy over a large range of altitudes and Mach numbers up to 20 in a real hypersonic flight environment.

APPROACH

The raw air data will be pneumatically sensed and enhanced (post-flight) using inertial navigation system (INS) data and global positioning system (GPS) data. The INS and GPS data will be used to filter and upgrade fidelity using minimum variance (Kalman Filter) estimation techniques.

VEHICLE CONFIGURATION

- No special hypersonic flight configuration is required to develop and validate the FADS.

INSTRUMENTATION

- six surface orifices distributed circumferentially on the conical surface located in or near stagnation region
- INS parameters, ground based radar, GPS, atmospheric data

ISSUES

- Surface pressure orifices located near the stagnation region may cause flow contamination downstream experiments.
- The ability to locate surface orifices near stagnation

tion region may be difficult due to requirements for cooling the leading edge or nosetip region.

Scramjet Engine Inlet/Combustor Operability and Performance JUSTIFICATION

Critical features of this system are sensitive to features of the true flight environment which cannot be fully simulated on the ground (such as real gas effects, quiet freestream flow, and scale effects). Flight data will provide an increased confidence level in the high Mach scramjet design process. The risk associated with scramjet performance would be reduced if flight data were available.

OBJECTIVE

To obtain scramjet engine operability and performance data in a high Mach number flight environment for comparison with ground test and CFD while maintaining objectives of inlet performance and operability.

APPROACH

Design an inlet, combustor, and partial nozzle scramjet, high Mach number flow path using the existing ground data base and CFD. Predict scramjet flow path performance with CFD analyses. Fabricate subscale and large scale models of the scramjet flow path and test in ground facilities simulating flight conditions. Fabricate large scale scramjet propulsion system for hypersonic test vehicle and flight test measuring propulsion flow path. Determine ground facility and CFD correlations to flight and improve CFD and ground test techniques.

VEHICLE CONFIGURATION

- Hypersonic flight test vehicle with deployable inlet to simulate actual flow path conditions in the flight environment.
- Fabricate largest scale scramjet feasible (sizing criteria based on ratio of boundary layer thickness at cowl lip to inlet height of 1/5).
- Inlet compression surfaces designed to provide the compression ratio and flow quality required for supersonic combustion.
- Inlet approach flow and inlet flow path must be as nearly 2-dimensional as possible.
- Test design must have geometric relevance to NASP-type configurations.
- No ablation or only minimal amount of ablation along conical surfaces upstream of inlet is allowed.
- The scramjet must have fixed geometry or if necessary, minimum variable geometry.

FLIGHT CONDITIONS

- scramjet testing at Mach numbers between 8 and 15
- constant Mach number and zero angle-of-attack to maintain cowl lip shock wave on body side shoulder

- 5-10 seconds test time at chosen flight Mach number
- near 2-dimensional, steady flow entering the inlet during test time
- fully turbulent boundary layer on vehicle surface

INSTRUMENTATION

- two pop-up and retractable pressure, temperature, and gas sampling rakes at each location of inlet entrance, throat, combustor entrance, combustor exit, and nozzle exit
- 50 to 100 static pressures on inlet ramps, cowl, combustor, and nozzle
- 10 to 20 thermocouples in combustor and nozzle
- non-intrusive instrumentation where possible
- accelerometers to measure force and moments

ISSUES

- Can survivable instrumentation be fabricated for the flight conditions?
- How accurate can flight conditions be determined (Mach, alpha, altitude, and dynamic pressure)?
- Will flight data accuracy be adequate to determine scramjet performance?
- Can a viable variable geometry capability be designed for a flight system?
- Is adequate cooling technology available to eliminate requirements for ablative surfaces ahead of the inlet?
- Can the cowl lip, combustor, and nozzle be designed to withstand flight conditions?
- Is adequate understanding of fuel mixing and fuel ignition systems available to design a functional scramjet engine?
- Can adequate control of pitching moment during scramjet burning be designed into the flight test vehicle?
- Is non-intrusive instrumentation capability adequately understood to utilize for flight test?

Very-High Temperature Ceramic Leading Edge and Nosetip Experiment JUSTIFICATION

The development of reusable, small radius leading edge and nosetip concepts which can survive hypersonic flight environments are essential for proposed hypersonic vehicles. Passive designs offer the simplest, lowest cost, and most reliable approach for developing these components, and also will provide a technology base which can be augmented with active cooling concepts to increase capabilities if needed. Flight testing of developmental leading edge and nosetip components is required to obtain the true thermal, pressure, and chemical environments, for evaluating material and component interactions, and to assess real at-

mospheric effects on performance.

OBJECTIVE

To evaluate the performance of very-high temperature ceramic leading edge and nosetip designs in a hypersonic flight environment.

APPROACH

New zirconium (Zr) and hafnium (Hf) based reinforced ceramic composites which show potential for very-high temperature reusability of 4000 degrees Fahrenheit) have been developed. Arcjet results and analysis predict that these ceramics could be used for small radius (less than 1 inch), non-ablating, passive nosetip and leading edge components on hypersonic vehicles to high Mach numbers (up to 10 at an altitude of 100K feet). Several passive nosetip and leading edge concepts incorporating these new very-high temperature reinforced ceramics would be analyzed and evaluated, and a selected nosetip and leading edge design would be fabricated and tested on a hypersonic flight vehicle. Instrumentation would include embedded thermal couples to monitor temperature, and possibly surface calorimeters and pressure measurements to establish heat flux and pressure loading. It would be highly desirable that the vehicle be recovered on land (and not immersed in salt water) for post-flight evaluation and characterization of the leading edge and nosetip components.

Surface Catalysis Experiment

JUSTIFICATION

Presently, the effect of surface catalysis on the heat transfer rate to the thermal protection system of future space vehicles such as Space Transfer Vehicle, Aeroassist Flight Vehicle, and National Aero-Space Plane are estimated using atom recombination coefficients obtained from ground base facilities. Low temperature coefficients are calculated from side-arm reactor data, using kinetic theory, and the high temperature coefficients are calculated from arc-jet data using aerothermodynamic theory. The actual thermochemical environment for high altitude hypersonic flight cannot be simulated accurately in ground test facilities.

OBJECTIVE

Verify analytical predictions that include the effects of surface catalysis on the convective heating rates to the thermal protection system (TPS) of a hypersonic flight test vehicle.

APPROACH

Apply coatings of known catalytic efficiencies (obtained from ground test facilities) to selected locations along the centerline of the vehicle. Obtain surface temperature jump and relaxation data due to a discrete catalytic area along the lower surface of the vehicle by using an array of thermocouples or calorimeters. Calorimeter or temperature measurements would also be required to de-

fine the heating distribution along the center line of the vehicle's heat shield during the flight.

Flight Performance of Advanced Thermal Protection Materials Experiment

JUSTIFICATION

The performance of advanced ceramic heat shields in hypersonic flight environments (where, high temperature real chemistry and pressure combined with handling) cannot be simulated in ground test facilities. Many types of hypersonic vehicles will require the use of non-ablative materials in order to perform their aerodynamic and propulsion experiments. The need for producing these materials and evaluating real flight effects on them has been recognized previously by NASA as evidenced by the Advanced Thermal Protection Materials experiments on the Orbiter Experiment (OEX) on the Space Shuttle and the Aeroassist Flight Experiment (AFE). In addition to allowing characterization of the materials in flight, the OEX studies accelerated use of several materials including LI-2200 and FRCI-12 rigid insulations.

OBJECTIVE

To evaluate the performance of advanced ceramic thermal protection materials in a hypersonic flight environment.

APPROACH

Advanced thermal protection materials will be applied in selected areas on the exterior of the SWERVE/Pegasus vehicle as a replacement the baseline heat shield. Appropriate instrumentation (such as thermocouples, pressure transducers) will be used to evaluate material response and performance in the flight environment. The materials to be considered for testing will include light weight, rigid and flexible insulations, high temperature structural ceramics, and very-high temperature ceramics (Zr and Hf ceramics).

ISSUES

- Because of maximum temperature and heat flux capabilities, the location and type of advanced ceramic heat shield materials to be used on the hypersonic test vehicle will be a function of the selected flight envelope and the programmed angle of attack.
- At large angles of attack, possible shock interactions away from the leading edges becomes an important design issue.

VI. PROJECTED HYPERSONIC FLIGHT EXPERIMENTS

A. Pegasus Piggy-Back Program

The Pegasus vehicle was developed by Orbital Sciences Corporation (OSC) for DARPA. Pegasus is a four stage, air-launched, vehicle capable of placing 600 pounds into a 250 nautical mile polar orbit or 1100 pounds into level flight at Mach

numbers approaching 20 and an altitude of 200K feet. A typical trajectory is shown in figure 2. This is more than adequate performance for a hypersonic flight research program. The vehicle is unique in that it uses aerodynamic lift in the initial part of the boost trajectory after release from the carrier airplane. It is in reality, a hypersonic airplane until first stage burn-out. The fact that it is an airplane provides a unique opportunity to study airplane-like phenomena during flight (ref. 4). Previous hypersonic flight experiments have focussed on conical bodies that are not representative of real hypersonic lifting configurations.

The first stage burns out at a Mach number of approximately 8.5. When the vehicle is used as a satellite launcher, the second and third stages continue to accelerate the payload into orbit. These stages are not suitable for carrying hypersonic flight experiments due to the low dynamic pressures they experience. The first stage contains sufficient volume to carry piggy-back experiments. The small additional weight from these experiments has minimal effect on the overall performance. All of our proposed piggy-back experiments fit within the first stage operational envelope.

Summary of Piggy-Back Experiments

Several discrete experiments have been selected as being of highest importance to the technology base and match well with the unique capabilities and design features of the Pegasus.

Pegasus Aerothermal Survey (Flight numbers 1 and 2)

These experiments are discussed at length in references 5 and 6.

Boundary Layer Transition, (figure 3), (T-S wave, crossflow, pressure gradient, and combinations with pressure gradients), (Flight numbers 4, 5, and 6)

The boundary layer transition experiment requires a special glove design which is discussed in references 7 and 8:

- wing root local flow and boundary layer survey
- a control gap heating assessment
- trajectory and maneuver derived performance coefficients
- an evaluation of various methods for deriving air data
- validation of hypersonic Reynolds number analogy for design codes
- development and evaluation of sensors for skin friction measurement
- boundary layer transition, and atmospheric particulate determination
- by inference, an assessment of atmospheric

conditions throughout the flight

These experiments provide data of a generic nature that will be applicable to a wide variety of vehicle types and will be useful for calibration and correlation with ground tests and analysis. In addition, some of them will be useful for validation of the Pegasus vehicle design itself. The initial flight of the Pegasus occurred in April, 1990. The piggy-back experiments are planned for inclusion on the initial six Pegasus flights. These flights are expected to be conducted in the 1993 to 1995 time period.

B. SWERVE/Pegasus Program

NASA has sponsored an extensive study effort by Sandia National Laboratories to formulate a hypersonic flight research program using the SWERVE vehicle. The SWERVE vehicle is configured (the shape is classified) for horizontal gliding flight from high hypersonic speeds down to Mach numbers of 2. Thus in one flight, data can be obtained over a very wide speed range including flight regimes where ground test capability exists. In addition, the vehicle is controllable; thereby enabling conventional airplane type maneuvers to be performed which further enhances data return and research options.

A series of dedicated flights were defined that use the Pegasus as the accelerator and the Sandia National Laboratory's SWERVE as the research vehicle. For these flights, the Pegasus is only the means to get the experiment to the proper flight conditions and is not the source of the research data. Vehicles other than SWERVE can be used but the SWERVE is mature, well characterized, and well suited for the intended research objectives. Cost and operational flexibility are the factors that make the Pegasus the launch vehicle of choice. Currently, funds are not available in the NASA Generic Hypersonic Program to conduct the Pegasus/SWERVE flight tests.

Hypersonic flight research may indeed be a beneficiary of the end of the Cold War. New developments promise much lower cost access to the desired flight conditions using a horde of available ex-military rockets. Typically, for research purposes, a small solid rocket booster system is the choice to accelerate light weight payloads into the lower hypersonic speed area (Mach numbers from 5 to 8) and surplus intercontinental ballistic missiles or the like would be used for the higher speeds and higher payloads. In order get to Mach numbers approaching 20, a booster such as a converted Polaris is required.

Experiments proposed for the SWERVE/Pegasus

are:

- hypersonic boundary layer transition including relaminarization (figures 4 and 5)
- real gas effects on vehicle aerodynamics and dynamics
- aerodynamic characteristics including corner flows, shock interference, vehicle control effectiveness, and air data measurements
- structures and materials evaluation include actively cooled nose concepts, surface quality effects, and boundary layer/materials interactions
- upper atmosphere characterization
- sensor and instrumentation development
- antenna performance assessment

The Pegasus/SWERVE combination is an appealing approach for conducting hypersonic flight research. The Pegasus has more than sufficient performance to accelerate a SWERVE to over Mach numbers of 20 at a very nominal cost. The SWERVE vehicle is a mature design which has been flown previously at lower Mach numbers. The SWERVE shape is well suited for these experiments and the internal volume and systems arrangement also supports experiments such as the ones listed above; however, the SWERVE is not the only vehicle that can be used. If other experiments or developments necessitate, another vehicle in place of the SWERVE that must be given careful consideration.

C. Air Breathing Propulsion Experiments

As the NASP program has progressed, it is becoming apparent that it is not now possible to accurately predict installed propulsion system performance at hypersonic speeds. This then leads to the question as to whether any meaningful small scale experiments can be conducted to address this technical issue. The NASP program has provided new technology developments and a variety of vehicles are being considered for new test bed capabilities. The combination of developments implies a new look at potential hypersonic air breathing propulsion flight research is appropriate.

VII. CONCLUSIONS

To develop the technologies needed to fully exploit the capability of hypersonic flight, advanced research, prediction methodologies, ground and flight test programs must be undertaken. Flight research in the high Mach number, high dynamic pressure flight regime is essential to the development of future operational hypersonic systems.

The piggy-back experiments identified for the Pegasus will enable acquisition of research data of

broad applicability and also will directly support the development of a critical commercial launch system. The piggy-back experiments will develop instrumentation packages for hypersonic data acquisition and will provide unique data of high value to designers and researchers for hypersonic vehicles. The data will also validate the vehicle sign and design methods, could identify areas of the vehicle in need of further improvement, and will enhance the understanding of vehicle capabilities and performance margins. The Pegasus could provide the nation with a low cost, reliable, and responsive access to space.

NASA's charter is to conduct research in areas of importance to the nation for military as well as civil purposes. An essential part of that charter is conducting research in the hypersonic flight regime. NASA has extensive ground test capability and computational methods development activity underway for hypersonic flight technologies. In order for the program to be complete, a flight research element must be added. The three-legged stool of prediction, ground test and flight test, must be reinforced. The flight experiments identified and discussed here-in, are critical elements in the development of future aeronautical capabilities for the nation.

The flight data return is of critical importance as benchmark information for understanding the fundamental physics of hypersonic flight as well verification of design methods. Research vehicles and instrumentation systems would be developed to enable NASA to carry out its research and development charter in the hypersonic flight regime.

The hypersonic flight test research program offers a unique opportunity to allow for classical engineering solutions to these critical technology development problems. A problem is presented and engineers can go off and solve the problem using a systematic approach applied to real flight vehicles and experiments. It is this methodical engineering work which will ultimately lead to the successful application of new technologies developed in the Generic Hypersonics Research Program.

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APPENDIX I

The 1988 Generic Hypersonic Flight Test Workshop

The NASA Generic Hypersonic flight research program was defined in June 1988 at the Flight Test Workshop held at the Dryden Flight Research Facility (DFRF). This workshop was attended by representatives from NASA, other government agencies, industry, and academia. The following statement was identified as the objective of the workshop:

"The scarcity of hypersonic flight data has generated a critical need for this type of information to validate technology in a number of disciplines including computational fluid dynamics, materials, structures, and propulsion. As a result, NASA Headquarters is organizing a Hypersonic Flight Test Workshop. It is scheduled for June 21-22, 1988, at the Dryden Flight Research Facility, Edwards Air Force Base, California. The objective of this workshop is to develop an effective plan for obtaining the critically needed validation data in a cost-efficient way. The workshop will be devoted to reviewing on-going or planned flight test programs to determine how those programs can be effectively used to obtain the necessary validation data".

The conclusions reached during the workshop regarding hypersonic flight testing are:

- The United States has many assets that can and should be exploited for hypersonic flight testing. Ground test facilities cannot duplicate simultaneously, the combined effects of temperature, pressure, flight loads, and high Mach numbers. Hence flight testing will be a crucial ingredient for the successful development and fielding of a hypersonic capability. A national flight test program can readily be implemented using the test-beds and launch vehicles described below:
 - It appears that a low cost version is clearly needed ($M=4$ to say 10-12, with small front-end payloads). This national program should be as standard as possible (flight vehicle, launch vehicle, telemetry). It is also certain that a complementary program will be needed that can test up to $M=20$ with higher weights for the more difficult technical and design issues. Piggy-back (cost-sharing) operations can enable the acquisition of useful flight data at very low cost and should be exploited whenever convenient.
 - Since ground testing is limited to about $M=10$ (10,000 ft/sec); flight testing from $M=10$ to $M=20$ is very crucial. Above $M=3.5$, no "clean-flow, quiet tunnel" capability exists;

flight above $M=3.5$ would thus provide unique data for validating CFD codes.

- Propulsion testing from $M=6$ to 20 is a very high priority (inlets, combustors, nozzles). In view of the numerous potential hypersonic applications for air-breathers the propulsion units would come in a wide range of sizes.
- Aero/Aerothermal testing up to $M=20$ is crucial for boundary layer transition, and turbulent mixing.
- Structural testing up to $M=20$ is much needed for nose tips, leading edges, catalyticity, and large high temperature structural panels.

Some of the potential test-beds and launch vehicles that were discussed include the NASP X-30, the Orbital Sciences Corporation/PEGASUS Launch System, the General Electric/ Scramjet Flight Demonstrator, the Sandia/SWERVE Research Vehicle, the NASA/HYFIRE Research Plan, the LTV/Advanced Scout Booster using strap-ons, the Lockheed Corporation/SR-71 RPV System, and a NASA Langley comprehensive summary of past proposed hypersonic flight research vehicles and experiments. These test-beds and launch systems can provide excellent starting points for flight experiments that will lead to an understanding of many of the technological problems of hypersonic flight.

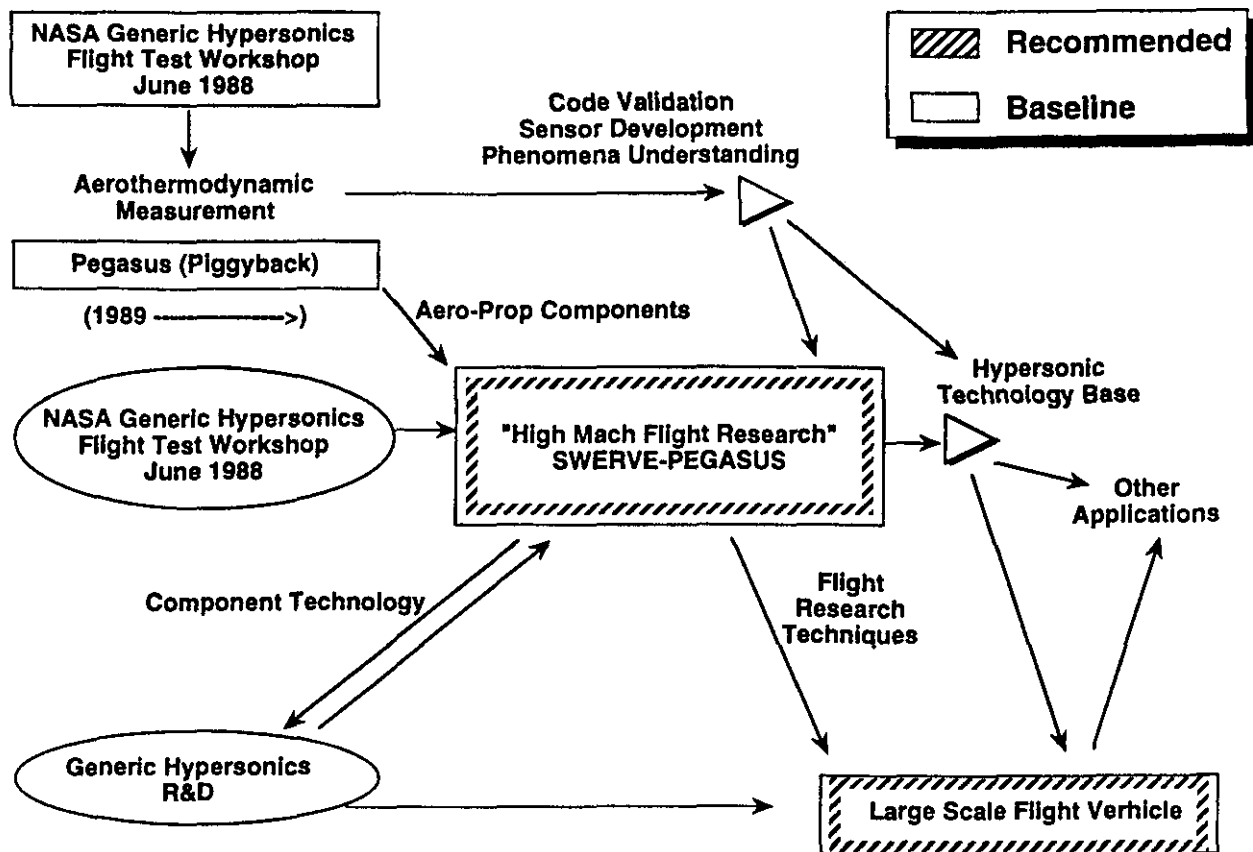


Fig. 1 Roadmap for Generic Hypersonics Flight Research

Fig 2 Hypersonic Flight Test Summary

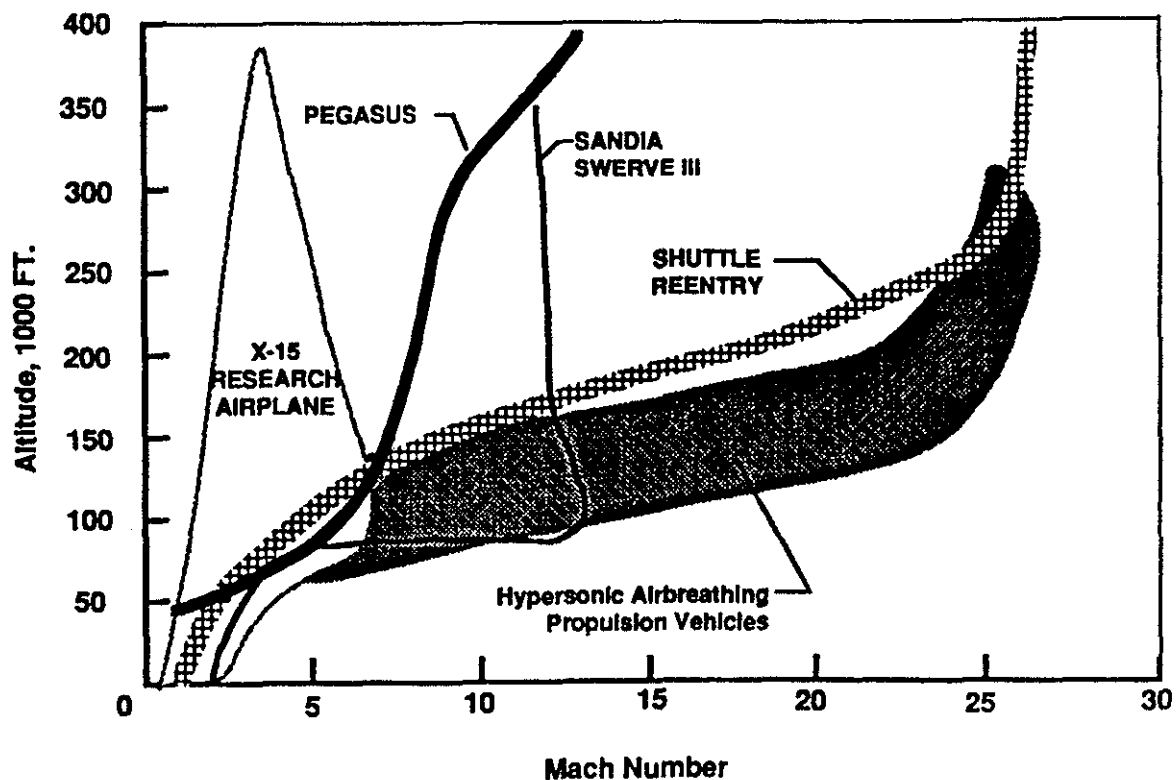


Fig 3 Pegasus "Add-On" Flight Research - Phase II

- 3 flight program, first flight in 1992
- Obtain cross-flow transition data at ideal gas Mach numbers
- Develop high frequency sensors and on-board spectral processing capabilities for hypersonic transition and turbulence studies

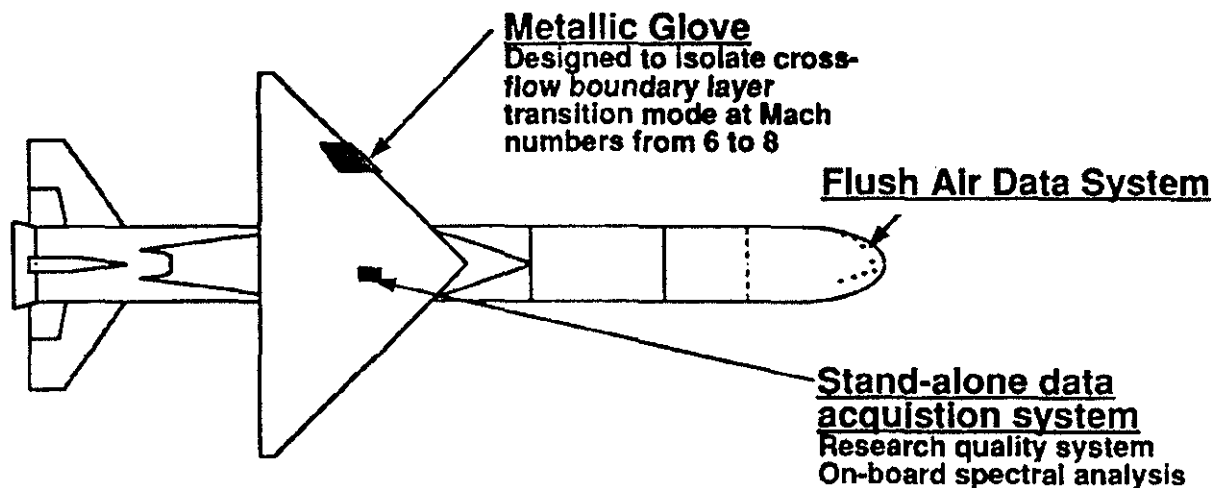
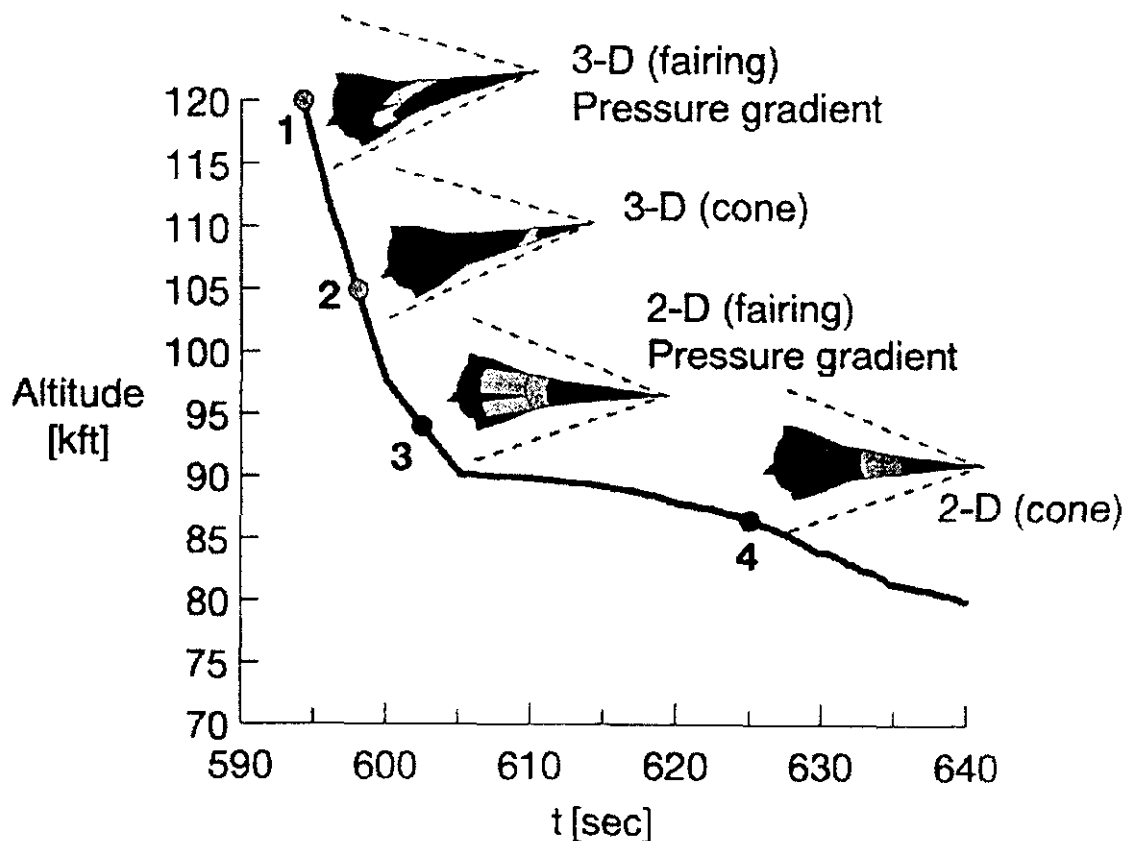
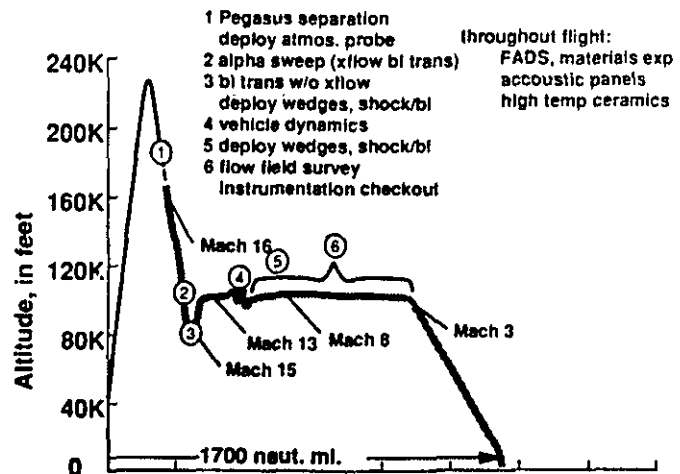


Fig 4 HYFLEX BOUNDARY LAYER TRANSITION EXPERIMENT



Flight #1

- 2-stage Pegasus
- Baseline 22 ft
- 0.4 inch nose
- FADS on conical surface
- Deployable wedges
- Pop-up rakes
- Materials and accoustic panels



Flight #2

- 2-stage Pegasus
- BI-conic or cone with sutable ramp
- 0.7 inch nose with cooling system
- Deployable device to promote transition ?
- FADS on nose and conical surface
- Deployable cowl-lip test
- Deployable inlet w/lip cooling
- Catalytic materials panels
- Pop-up rakes
- Materials and accoustic panels

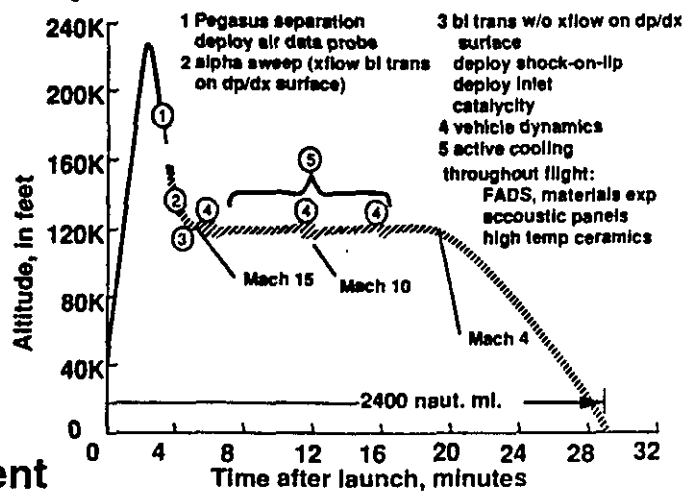


Fig. 5 HYFLEX Experiment